WI-FI SYNCHRONISED ANALOGUE CLOCK

WiFi Clock v2.0

User manual for the **latest Chouchin-CH899 board revision**

For movements fitted with HC32 V21 and TXW813 replacement firmware. This is not the manual for the older MM32/ESP board.

What you need
Two AA batteries suitable for your movement, a 2.4 GHz Wi-Fi network and a phone or computer. NTP supplies accurate time; the clock normally synchronises at its daily update time and powers down Wi-Fi between sessions.

Install the movement and hands

1 **Set the reference, then fit the hands**
If no transport pin is fitted, power the movement and hold **RESET for three seconds**. Wait until all hands reach 12, then remove power. Fit the hour, minute and optional seconds hand pointing exactly to 12. Keep hands clear of each other, the dial and glass.

2 **Remove the transport locking pin**
If supplied, leave the locking pin fitted while mounting and aligning the hands. Remove it before inserting batteries or operating the movement. Never force the hands or run the movement with the pin fitted.

3 **Power the movement**
Insert both batteries with the correct polarity. Allow calibration and time-setting to finish before assessing hand alignment. Check your movement's mounting dimensions; old-board drawings are not specifications for this revision.

First-time Wi-Fi setup

1 **Join WiFi-Clock-Setup**
With no saved settings, the clock provides this setup network. Connect your phone or computer; if warned that there is no Internet, stay connected. Open **http://192.168.4.1/** in a browser.

2 **Choose your network and clock settings**
Select your 2.4 GHz network and enter its password. Set timezone, daylight saving and daily update time. The two default NTP servers are suitable for normal use. Hand movement and battery-saving options are explained on page 2.

3 **Select Save settings**
The clock saves the configuration, reconnects to Wi-Fi and sends time to the movement after NTP synchronises. The page may become unavailable. Rejoin your normal network and use the clock's current router-assigned IP address.

No external website is needed for setup. The page is stored in the clock. Internet access is needed only when using an Internet-based NTP server.

Webpage settings

Open the page while the Wi-Fi chip is awake. The connection status shows whether it is in setup mode, connected or synchronised. A saved-setting confirmation does not by itself confirm successful NTP.

Wi-Fi network	**Available Wi-Fi networks** shows the saved network name, or **Select a network** if none is saved. Choose **Manual** to reveal **Manually enter Network name (SSID)** for a hidden or unlisted network.
Password	Use a 2.4 GHz open or WPA2-PSK network. Keep an existing password unchanged to retain it; saved passwords are not displayed. Enterprise Wi-Fi is not supported.
Time servers	Primary: **pool.ntp.org**. Secondary: **time.cloudflare.com**. After five seconds, the secondary is also tried while the primary remains eligible. The first valid reply wins. Hostnames or IPv4 addresses may be entered; blank fields use the defaults.
Timezone / daylight saving	Select the correct timezone. **Automatic** applies its seasonal rule; **Disabled** keeps standard time year-round. **Custom rule** displays the Custom POSIX rule box for an advanced rule.
Next DST change	Updates from your current selections after the clock has synchronised, without needing a page reload. Previewing does not save settings. A special DST wake does not replace your saved daily update time; Disabled schedules no special DST wake.
Daily update time	Any hour, with minutes **00, 10, 20, 30, 40 or 50**, in the clock's local time. The controller also supports minute-resolution special DST wakes. Normal daily settings remain in ten-minute steps.

Hand movement

Sweep	Seconds move continuously; the minute hand advances gradually.
Burst	Seconds move continuously; the minute hand advances in a jump.
Hold&Start	Replicates Swiss railway station-clock-style movement: the seconds hand pauses at 12, then restarts as the minute hand jumps forward. No official association or endorsement is implied.

Second hand battery saver

Off (default)	Normal selected movement, without scheduled seconds-hand parking.
Night parking	The seconds hand reaches 12 and parks between **00:00 and 06:00** to reduce battery use.
On	The seconds hand reaches 12 and stays parked indefinitely. Use this when no second hand is fitted to prolong battery life. The minute and hour hands continue to keep time.

Defaults: **Sweep** and battery saver **Off**. Select **Save settings** after changes. Original controller battery-protection checks remain active.

Daily use, recovery and care

> **Opening the page again**
> Find the awake clock in your router's client list and open its IP address. Its hostname is **WiFi-Clock-ABCDEF**, using the last six MAC hex digits in capitals. Hostname lookup depends on your router; **.local is not guaranteed**.

Hold **REC for three seconds** to wake Wi-Fi. With saved credentials, connect your phone or computer to the same LAN and open the clock's IP address. With no saved credentials, join **WiFi-Clock-Setup** and open 192.168.4.1. REC does not erase settings.

Keep the webpage open to hold Wi-Fi power on; close it afterwards for low-power operation. Firmware updates are wired, not over-the-air.

Erase saved settings

Hold **REC and RESET together for three seconds** to erase saved settings. Alternatively, open **Reset settings** on the webpage, choose **Erase settings** and confirm. Wi-Fi and clock settings are cleared; the setup network returns with Sweep and battery saver Off. This does not restore original firmware or erase factory radio data.

Quick troubleshooting

Cannot find setup Wi-Fi	A configured clock normally joins its saved network and may be asleep. Request setup access, or wait for a wake. Use the router-assigned IP when it is connected to your normal network.
Page disappears after saving	Reconnect your phone to the appropriate network. The clock may have changed between setup and normal Wi-Fi, received a different IP, or powered down. Open it again during an awake session.
Clock does not synchronise	Check Wi-Fi and live status, then the two NTP servers. The router must allow DNS and NTP (UDP port 123). An IPv4 NTP address can help diagnose a DNS issue; use a trusted server.
Seconds stay at 12	Check battery saver and movement style: On parks continuously, Night parking parks overnight, and Hold&Start pauses briefly each minute. A low battery can also activate the original protection behavior.
Hands are misaligned	Hold **RESET for three seconds** and wait until all hands reach 12. Remove power, then refit them pointing exactly to 12. Check clearance and remove any transport pin before powering on. RESET alone does not erase settings.

Safety and support

Batteries: use a matching pair with correct polarity; do not mix ages, chemistries or rechargeable/non-rechargeable cells. Remove exhausted or leaking batteries. **Care:** keep indoors and dry; do not alter the antenna or supply wiring. Recycle batteries and the movement locally.

Privacy: the setup network is open and the webpage uses local HTTP, not HTTPS. Use a trusted network; do not expose the page to the Internet. Saved settings are not encrypted. This manual is not a declaration of conformity or a measured battery-life specification.

Project guide and updates: github.com/maddenste/Chouchin-899-latest-revision
Project-authored work is GPL-3.0-or-later; vendor firmware rights are separate. The manufacturer is not affiliated with this project. Download availability is subject to publication and licensing clearance.

WiFi Clock v2.0 - Modified by Steve Madden - 2026
Manual updated 18 October 2026. Retain this guide with the movement.